

Real-Time Traffic Analytics Using Azure Stream Analytics

Project Proposal – DEPI Data Engineering Track

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Executive Summary

This project aims to design and implement a real-time traffic analytics system using Microsoft Azure Stream Analytics. The system simulates vehicle traffic data, ingests it into Azure services, processes it in real-time, and provides dashboards and alerts for decision-making. The project enables the team to gain practical experience with cloud-based streaming data pipelines, data processing, visualization, and alerting systems.

Project Overview & Objectives

The project introduces students to real-time data streaming, analytics, and visualization. By simulating traffic data and processing it with Azure Stream Analytics, the project provides a framework to analyze traffic congestion, detect anomalies, and deliver insights through dashboards and alerts.

Objectives:

1. Simulate and ingest traffic data into Azure Event Hubs or Blob Storage.
2. Process and analyze data streams in real-time using Azure Stream Analytics.
3. Store processed insights in Azure SQL or Data Lake.
4. Visualize results through dashboards and trigger alerts when thresholds are exceeded.

Milestones, Tasks, and Deliverables

Milestone 1: Traffic Data Simulation and Ingestion

Objectives:

- Generate and stream traffic data into Azure Event Hubs or Blob Storage

Tasks:

1. Simulation Script: Create a Python script to simulate vehicle count, speed, timestamp, location
2. Azure Setup: Push data to Azure Event Hubs or Blob Storage

Deliverables:

- Python data generator
- Azure Event Hub with data streams

Milestone 2: Real-Time Processing

Objectives:

- Analyze traffic patterns and detect anomalies

Tasks:

1. Azure Stream Analytics: Create real-time queries to detect congestion, high-speed vehicles, or accidents
2. Output: Store insights in Azure SQL or Data Lake

Deliverables:

- Real-time query configurations
- Database or data lake with processed insights

Milestone 3: Dashboard and Alerting

Objectives:

- Visualize live traffic data and trigger alerts

Tasks:

1. Visualization: Use Power BI or a Streamlit dashboard to visualize traffic
2. Alerts: Send alerts when traffic exceeds defined thresholds

Deliverables:

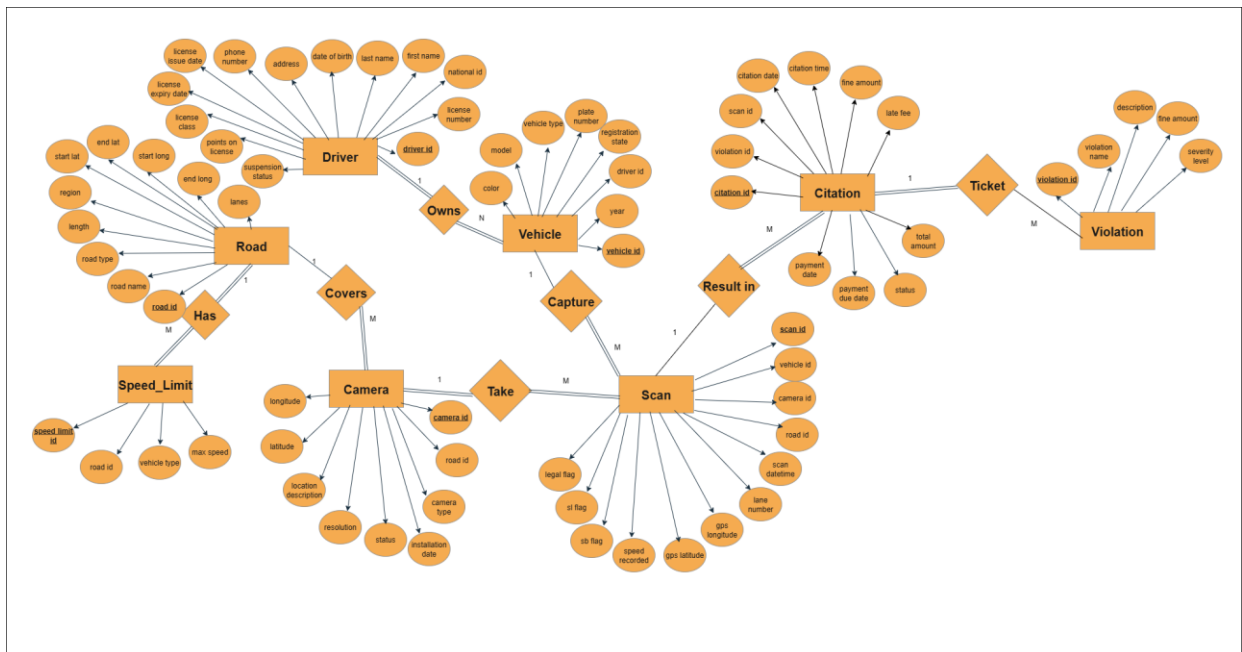
- Live dashboard screenshots
- Alert log or notification system
- Final report summarizing system architecture

System Design

The system design is represented through the following Entity-Relationship Diagram (ERD) and database schema.

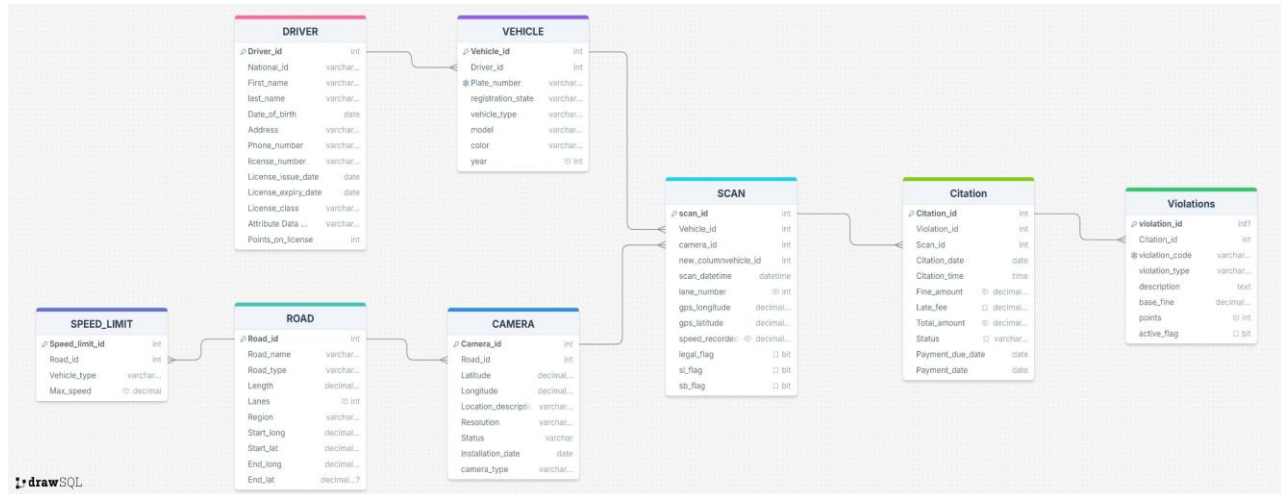
Entity-Relationship Diagram (ERD)

The following diagram illustrates the ERD for the project.



Database Schema

The following schema represents the database structure for storing traffic insights and related data.



Implementation Plan

The project will be implemented using Microsoft Azure cloud services and supporting tools as follows:

- Data Simulation: Python script generating traffic data.
- Data Ingestion: Azure Event Hubs or Blob Storage.
- Real-Time Processing: Azure Stream Analytics queries.
- Data Storage: Azure SQL Database or Data Lake.
- Visualization: Power BI dashboards or Streamlit web application.
- Alerting: Threshold-based notifications using Azure services.

The team will follow a milestone-driven approach to ensure timely delivery of each component.

Expected Outcomes & Benefits

Upon completion, the project will deliver a fully functional real-time traffic analytics system. This system will:

- Provide hands-on experience with Azure Stream Analytics and data engineering practices.
- Demonstrate how real-time insights can support traffic management and decision-making.
- Serve as a reference architecture for similar IoT and streaming data analytics projects.